AP & RAP Assignment 5

1. Let X be a random variable with $X \geq 0$. Prove that

$$\lim_{y\to\infty} yE(1/X; X > y) = 0, \qquad \lim_{y\downarrow 0} yE(1/X; X > y) = 0.$$

2. Let X and Y be random variables. Prove that if $p, q \in (1, \infty)$ satisfy $1/p + 1/q = 1$, then

$$E|XY| \leq \|X\|_p \|Y\|_q,$$

where $\|X\|_r = (E|X|^r)^{1/r}$, $r \in (1, \infty)$.

3. Let $(X_n : n \in \mathbb{N})$ be independent exponentially distributed random variables, i.e.,

$$P(X_n > x) = e^{-x}, \qquad x \geq 0.$$

Then for $\alpha > 0$

$$P(X_n > \alpha \log n) = n^{-\alpha}.$$

Let $A_n := \{X_n > \alpha \log n\}$, please prove:

$$P(\limsup_{n\to\infty} A_n) = \begin{cases} 0 & \text{if } \alpha > 1, \\ 1 & \text{if } \alpha \leq 1. \end{cases}$$

4. Let $A_1, A_2, \cdots$ be a sequence of independent events, and for all $n \in \mathbb{N}$, we have $P(A_n) < 1$. Prove: if $P(\cup_{n=1}^{\infty} A_n) = 1$, then $P(\limsup_{n\to\infty} A_n) = 1$.